METHODOLOGY REPORT

Bank & NBFC Risk Workbench

A comparative financial distress and structural credit-risk system for Indian banks and NBFCs. How both analyses are built, how they were tested against real RBI actions and defaults, and what they cannot tell you.

Universe	556 Indian banks and NBFCs (39 banks, 517 finance companies)
Panel	6,223 company-years, 563 institutions, FY2006-FY2023 (dense from FY2012)
Accounts	Detailed Financials of 4,456 NSE & BSE companies (public archive)
Prices	303,995 daily closes, 180 symbols, 2,713 trading days, from NSE's own archive
Analysis 1	Fundamental Distress Index - reads the published accounts
Analysis 2	Merton / KMV distance-to-default - reads the equity market
Validation	18 labelled distress events, each with a public source
Headline	7 of 7 held-out events caught by a rule that never saw them, p = 0.0003
Coverage	18 of 22 known distress events now evaluable, after recovering 7 from source

Academic decision support. Not investment advice and not a credit rating.

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What is in this report

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III - Low level: Analysis 2	The Merton solver, the price pipeline, and the corporate-action problem that had to be solved exactly.	Reproducing it
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PART I

High-Level Explanation

The tool reads published accounts and market prices and produces a ranked view of which lenders look fragile. It is not forecasting a date; it is answering comparative questions, and showing where two independent ways of asking disagree.

1. The problem

When a bank or NBFC fails, the failure is announced on a single day - a moratorium, a default, a regulator superseding the board. But the deterioration that caused it took years, and most of it sat in public filings the whole time. Credit ratings and regulatory action tend to confirm distress rather than anticipate it. The question asked each year, for each lender, is: **compared with institutions like it, how bad does this look, and is it getting worse?**

2. Two analyses, deliberately different

The tool runs two analyses that read different evidence. That is the point of the design, not a convenience:

	Analysis 1 - Fundamental	Analysis 2 - Structural (Merton)
Reads	Published annual accounts	Daily equity prices and debt
Technique	Financial statement analysis	Option pricing (Black-Scholes)
Output	Distress index, 0-100	Distance to default, implied PD
Covers	All 556 institutions	152 listed with a liquid quote
Strength	Works for unlisted firms; directly interpretable	Moves daily; prices in what investors suspect but accounts do not yet show
Weakness	Inherits whatever the accounts conceal	Needs a market; assumes no parent or state support

The weakness of the first is exactly the strength of the second. Yes Bank's reported profitability ranked mid-pack right up to the end, because the bad loans had not been recognised. The share price had already collapsed. A third page in the tool is built on the gap between the two.

3. The data

The accounting archive holds 4,492 listed Indian companies, each a folder of eight CSV files covering balance sheet, profit and loss, cash flow, ratios and shareholding. The 556 institutions whose sector is Banks or Finance are kept, giving 6,139 company-years from 2006 to 2023, dense from about 2012.

Two inconsistencies had to be resolved first. The files use two date formats in their column headers. More importantly, lenders file their profit and loss in two different shapes: 75 report *Revenue* and *Financing Profit*, the other 481 report *Sales* and *Operating Profit*. These are the same concepts under different labels; without unifying them 87% of the panel would carry a blank revenue column.

Prices come from NSE's own daily bhavcopy archive rather than yfinance, which is covered in Part III.

4. What Analysis 1 measures

Six readings per institution per year, each a different route into trouble:

- **How thick the cushion is.** Equity as a share of total assets - the money that absorbs losses before depositors and lenders start losing theirs.
- **Whether the cushion is shrinking.** The two-year change in equity. This is the reading that caught Reliance Capital: reserves fell from Rs 15,081 crore in FY2017 to Rs 760 crore in FY2018, then turned negative. The collapse came in late 2021.
- **Whether it earns money.** Return on assets. A lender that cannot earn its way out of a problem must shrink or raise capital, and distressed firms can rarely do either.
- **How much it borrows against that cushion.** Borrowings divided by equity.
- **How much income interest consumes.** A lender paying most of its revenue to its own creditors has little margin for error.
- **Whether earnings are trending down.** Return on assets against three years ago, which catches slow bleeds a single snapshot makes look acceptable.

5. Comparing like with like

A raw ratio means nothing in isolation. Seven percent equity-to-assets is ordinary for a bank and alarming for a finance company. So every reading becomes a rank - where this institution sits among comparable ones that year - on a 0 to 100 scale where 100 is always worst. Ranking self-calibrates to the cycle: in a bad year absolute numbers worsen but the ranking still identifies who is weakest, with no fixed thresholds to go stale.

The first attempt was wrong, and the error is instructive. Ranking all 556 institutions together put HDFC Bank at 66 against a sector median of 50 - the model was calling one of the safest banks in the country above-average risky. Most of the 517-company Finance sector are not lenders at all but small investment and holding companies carrying almost no borrowings. Against that crowd, any institution funding itself normally - the whole business model of banking - looks dangerously leveraged.

Comparison sets were narrowed: banks against the other 38 banks, NBFCs against NBFCs of similar size. HDFC Bank fell to the 30s and stayed there.

That fix created the opposite problem. If an entire group is uniformly thin on capital, ranking inside the group makes them all look fine by their own standards - precisely the gap separating the weak public-sector banks from the private ones, which sat at 5-6% equity against 10-14%. So the two capital readings are scored twice, once against peers and once against the whole sector, and averaged. Measured on the eleven events available at the time, this moved accuracy from 0.859 to 0.912 and took the model from catching 6 of those 11 to 9.

6. Why the alert has two triggers

The obvious rule is to alert when the score is high. That rule fails on the most important case in the dataset. Yes Bank's FY2019 accounts, public June 2019 and nine months before the moratorium, scored **69** - not low, but not high enough to flag. Yet 69 was wildly out of character: its own scores over the preceding years were 44, 40, 40, 45, and its three-year median was 39.9. FY2019 was a **29-point jump against itself**.

The rule became: alert if the score is 80 or above, **or** if it is above 60 and has risen 15 or more points above the institution's own recent median. A high score tells you *who* is weak; a rising score tells you *when* something changed.

7. What Analysis 2 measures

Merton's insight is that owning equity in a leveraged firm is the same payoff as owning a call option on its assets, struck at the face value of its debt. If assets end above the debt, shareholders keep the difference; if below, they get nothing and the option expires worthless. That makes the Black-Scholes machinery from the derivatives part of the

course directly applicable to credit risk.

Asset value and asset volatility are not observable, but equity value and equity volatility are. Two equations - the option price and its volatility relationship - are solved for the two unknowns. The result is **distance to default**: how many standard deviations of asset value sit between the firm and its debt. Feeding that through the normal distribution gives a model-implied probability of default.

The number this produces is an output of the model's assumptions, not an observed default rate. Its value is comparative and directional: that Yes Bank's implied probability rose from 1.5% to 36% over fifteen months is far more meaningful than the 36% itself.

8. The tool

A Streamlit application with a menu of five pages:

Page	What the user does
Data Explorer	Filter by sector, peer group, year range and institution, then compute descriptive statistics on that selection - mean, median, standard deviation, quartiles, IQR, skew, correlation matrix, year-on-year changes, distribution and data coverage. Nothing is precomputed; results download as CSV.
1 - Fundamental Distress	Pick an institution and year. See the six components with raw values, percentiles and weighted contributions, the trajectory against the peer median, the alert status and the full peer ranking.
2 - Structural Credit Risk	Pick a listed institution. See distance to default and implied PD over time, plus the model inputs. Risk-free rate, horizon and default point are sliders, so the user can test how far the result depends on the assumptions.
3 - Comparative Assessment	Both analyses on common percentile axes for a chosen year, with the institutions where they disagree most listed in both directions.
Backtest & Validation	Lead times against the real events, distressed names against healthy controls, and why accuracy alone is the wrong test.

9. The finding worth arguing about

On the standard accuracy measure, **a single leverage ratio beats the entire fundamental model** - 0.976 against 0.900. Taken at face value that makes the project pointless. It is not, for two reasons. First, seven of the eleven events are thinly capitalised public-sector banks put under Prompt Corrective Action in 2015-17, so any ratio that effectively detects “is this a PSU bank” scores brilliantly without predicting anything. Second, and more damning, **leverage alone never changes its mind**: at a matched alert rate it flags 22 institutions continuously for eight years or more - HDFC Bank, ICICI, Axis, Canara and Bank of Baroda among them - and not one of the institutions it permanently flags ever failed. A smoke detector that is always sounding has perfect recall and no value.

PART II

Low Level: Fundamental Distress Index

Exact definitions, transforms, thresholds and edge cases, in pipeline order. Implemented in build_panel.py and score.py; reruns from the archive with no network access.

10. Data ingestion

- **Sector filter.** Keep folders whose Basic_Info sector is `Banks` or `Finance`: 39 and 517 respectively.
- **Column parsing.** Headers appear as `2019-03-01` or `Mar 2019`. Only March columns are retained, so the September 2023 interim column is dropped as a partial period.
- **Schema unification.** `revenue` takes Revenue or Sales; `op_profit` takes Financing Profit or Operating Profit. First non-null wins.
- **Share count inputs.** Face value is carried from Basic_Info so share counts can be derived per year rather than from a present-day snapshot. Part III uses this.
- **Asset quality.** Gross and Net NPA exist only on the quarterly P&L and only for recent quarters. They are carried as current context and never used in the backtest.

11. Feature definitions

```
equity            = Equity Capital + Reserves

f_capital_buffer  = equity / total_assets
f_capital_erosion = (equity[t] - equity[t-2]) / total_assets[t]
f_leverage        = borrowings / equity          (equity > 0 only)
f_roa             = net_profit / total_assets
f_earnings_trend  = f_roa[t] - f_roa[t-3]
f_interest_burden = interest_exp / revenue
```

Lags are taken within company after sorting by fiscal year, so a gap in a company's history shifts the lag rather than silently comparing non-adjacent years. Total assets or revenue of zero become null rather than infinity.

12. Peer-group construction

```
if sector == 'Banks':   group = 'Banks'                  (39 institutions)
else:                   group = 'NBFC-' + size_tercile

size_tercile = qcut( rank(total_assets) , 3 ) within each fiscal year
               -> Small | Mid | Large
```

Group sizes in FY2019: Banks 39, NBFC-Small 168, NBFC-Mid 167, NBFC-Large 167. Terciles are recomputed annually, so a growing NBFC can change group.

13. Percentile transform

Each feature is first oriented so a larger number always means more risk (capital buffer, capital erosion, ROA and earnings trend are negated), then ranked within its (fiscal year x peer group) cell:

```
s_<component> = rank(oriented_value, pct=True) * 100     -> 0 .. 100

nulls stay null (not imputed, and they do not take a rank)
cells with fewer than MIN_PEERS = 20 institutions are left unscored
```

14. Dual anchoring on capital

```
DUAL_ANCHOR = { capital_buffer , leverage }

s_component = mean( rank within (fy, peer_group) ,
                    rank within (fy)             )
```

15. Edge cases

- **Negative or zero equity.** The firm is balance-sheet insolvent and no percentile is meaningful. Both capital components are forced to 100 and the leverage ratio is left undefined rather than reported as a negative multiple.
- **Missing components.** The weighted mean is renormalised over whichever components are present.
- **Insufficient coverage.** Rows with under 60% of components available are left unscored. 103 of 6,139 rows fall out this way.

16. Aggregation and alert rule

```
WEIGHTS = { capital_buffer  : 0.30 ,  leverage        : 0.12 ,
            capital_erosion : 0.20 ,  interest_burden : 0.10 ,
            profitability   : 0.20 ,  earnings_trend  : 0.08 }

FDI = sum( s_c * w_c for available c ) / sum( w_c for available c )
FDI = null  if coverage < 0.60
band: Low < 60 | Watch 60-80 | Elevated 80-90 | High ≥ 90

fdi_base3 = median of the previous 3 fiscal years' FDI
            ( shift(1) then rolling(3, min_periods=2).median() )
alert = ( FDI ≥ 80 ) OR ( FDI ≥ 60 AND FDI - fdi_base3 ≥ +15 )
```

The baseline is shifted one year before the rolling median, so the current year never contributes to the baseline it is judged against.

17. Point-in-time discipline and materiality

```
PUBLICATION_LAG_MONTHS = 3
known_at(FY) = 31-Mar-FY + 3 months  =  30-Jun-FY
HORIZON_MONTHS = 24
is_pre_event = 0 < (event_date - known_at) ≤ 24 months
post_event   = event_date ≤ known_at   -> excluded from evaluation

is_material  = ( total_assets ≥ 500 cr ) AND ( revenue ≥ 25 cr )
FY2023: 151 material of 552 scored institutions
```

Roughly a third of the Finance sector is dormant shells: no book, no income, often deeply negative equity. The score correctly calls them insolvent and they swamp any ranked screen, so they are flagged rather than dropped.

18. Worked example: Yes Bank, FY2019

Accounts public 30 June 2019; moratorium 5 March 2020. Ranked against 39 banks. All figures in Rs crore.

```
equity   =     463 + 26,424 =  26,887
assets   =                   380,860
ROA      =   1,709 / 380,860 =  0.0045
leverage = 335,982 / 26,887  =  12.50
erosion  = (26,887 - 22,039) / 380,860 = 0.0127   [FY2017 equity 22,039]
trend    = 0.00449 - 0.01531 = -0.0108            [FY2016 ROA 0.01531]
```

Component	Raw	Percentile	Weight	Contribution
Capital buffer (thin capital)	0.0706	78.7	0.30	23.62
Capital erosion	0.0127	57.1	0.20	11.43
Profitability (ROA)	0.0045	51.3	0.20	10.26
Leverage	12.50	80.4	0.12	9.65
Interest burden	0.6687	71.8	0.10	7.18
Earnings trend	-0.0108	85.3	0.08	6.82
Fundamental Distress Index				**68.96**

69 is below the level threshold of 80, so a level-only screen stays silent. Yes Bank's trailing three-year median was **39.9**, making FY2019 a **+29.0** move against itself, and the deterioration leg fires **8.2 months before the moratorium**. Note what the accounts did not reveal: profitability still ranked at the 51st percentile.

19. Worked example: Reliance Capital, capital erosion

Fiscal year	Reserves (Rs cr)	Equity (Rs cr)	Equity / assets	FDI	Alert
FY2016	13,890	14,143	0.211	65.0	no
FY2017	15,081	15,334	0.186	70.2	no
FY2018	760	1,013	0.012	92.1	**yes**
FY2019	-1,202	-949	-0.011	90.1	yes
FY2020	-2,547	-2,294	-0.035	84.4	yes
FY2021	-11,791	-11,538	-0.178	92.9	yes

FY2018 accounts became public 30 June 2018, **41 months** before RBI superseded the board. From FY2019 the negative-equity rule pins both capital components at 100.

PART III

Low Level: Merton Structural Credit Risk

The option-pricing analysis, the price pipeline behind it, and the corporate-action problem that had to be solved exactly rather than approximately.

20. The model

```
E  = V*N(d1) - D*exp(-r*T)*N(d2)
sE = (V/E) * N(d1) * sV

d1 = [ ln(V/D) + (r + sV^2/2)*T ] / ( sV*sqrt(T) )
d2 = d1 - sV*sqrt(T)

   E  = market value of equity        V  = market value of assets
   sE = equity volatility             sV = asset volatility
   D  = default point (debt)          T  = horizon, 1 year

DD = [ ln(V/D) + (r - sV^2/2)*T ] / ( sV*sqrt(T) )
PD = N(-DD)
```

E and sE are observable; V and sV are not. The two equations are solved simultaneously by fixed-point iteration, the standard KMV approach: start at V = E + D and sV = sE*E/(E+D), substitute repeatedly until both move by less than 1e-6, capped at 300 iterations. Non-convergence returns null rather than a partial answer - which happens mostly when equity is tiny against debt, itself an informative state.

21. Why not yfinance

The project brief suggests yfinance. Yahoo's endpoint IP-throttled this machine with HTTP 429 on every attempt across roughly 45 tries, and the usual fallback source gates behind a bot-detection challenge, which is not something to work around.

NSE publishes its **daily bhavcopy** as plain static ZIP files - the primary source Yahoo itself derives from. One file holds every symbol traded that day, so a single request covers all 180 listed institutions in the panel at once. The full 2013-2023 archive is **303,995 daily closes across 180 symbols over 2,713 trading days**. Volatility uses 250-day windows scaled by sqrt(252).

An earlier version of this pipeline sampled weekly to save requests. That turned out to be a false economy: weekly sampling made corporate actions impossible to identify reliably, which is the subject of the next section.

22. The corporate-action problem

The first run produced nonsense. Yes Bank showed 170% equity volatility through mid-2018 and an implied default probability of 68% at a time when it traded at Rs 170 and was considered healthy. HDFC Bank's price appeared to halve in September 2019.

Both were **missed stock splits** - Yes Bank 5:1 in 2017, HDFC Bank 1:2 in 2019 - sitting in the return series as fake 80% and 50% crashes and poisoning the rolling volatility for a full year afterwards.

The obvious fix is unsafe. Treating any large drop whose ratio is near a round number as a split would have deleted a real event: Yes Bank genuinely halved in a single week in 2019 on real news. A loose heuristic cannot separate the two cases, and getting it wrong erases exactly the collapse the project exists to detect.

A first attempt at an exact test was based on a false premise. It assumed NSE's PREVCLOSE field was already adjusted for corporate actions, so that CLOSE/PREVCLOSE would give a clean return across a split. It is not adjusted. On Yes Bank's split day the file reports CLOSE 377.05 against PREVCLOSE 1880.10 - the raw drop. Every computed factor collapsed to 1 and the detector returned nothing but noise from illiquid symbols. The weekly sampling was abandoned and the full daily series collected instead.

What actually works is the single-day ratio on the daily series. Because the opening price is adjusted mechanically on the ex-date, a corporate action lands almost exactly on a conventional fraction, while a crash does not:

```
Yes Bank   21 Sep 2017:   377.05 / 1880.10 = 0.2005   (1/5, 0.25% off)
HDFC Bank  19 Sep 2019:  1101.05 / 2187.75 = 0.5033   (1/2, 0.66% off)

accept if the ratio is within 1.5% of 1/k or k, for
k in [1.5, 2, 2.5, 3, 4, 5, 6, 8, 10, 20]
```

Three guards were added, each after inspecting a real false positive in the output:

Guard	Threshold	The false positive it removed
Exclude the 20% circuit band	1.25 removed from the ratio list	0.80 is exactly India's lower circuit limit, so locked-down days land on it precisely. It was labelling three of Reliance Capital's 2019 crash days as bonus issues.
Market breadth	reject days where >6% of symbols moved >10%	19 March 2020 had 12.5% breadth and was reading Indiabulls Housing's COVID crash as a 1.5x split. Genuine split days show 1.4-3.4% breadth.
Minimum history	60 prior trading days	Ujjivan Small Finance Bank's listing-day pop.
No clustering	drop pairs within 30 days	Two opposite 1.5x 'actions' five days apart in an illiquid micro-cap.

The result is **12 corporate actions** across the panel, all of which check out against the public record: Axis 5:1 (2014), Bank of Baroda 5:1 (2015), ICICI 5:1 (2014), Kotak 1:1 bonus (2015), Karur Vysya 5:1 (2016), REC 1:1 bonus (2016), Yes Bank 5:1 (2017), Cholamandalam 5:1 (2019), HDFC Bank 1:2 (2019), and three smaller names.

23. Market capitalisation and share count

Splits and bonus issues both halve the price, but they do different things to the share count, so they cannot be handled alike. A split cuts the face value and leaves equity capital untouched. A bonus capitalises reserves, so equity capital rises with the share count. Since the archive reports equity capital annually, the two are told apart by checking whether equity capital jumped across the action date.

The two uses of the price series then need different inputs:

```
VOLATILITY  needs clean returns, so it uses prices adjusted for every
            mechanical action - splits, bonuses and rights alike.

MARKET CAP  needs the true value of equity on the day:

   face_value[t] = face_value_current * product of SPLIT factors after t
   shares[t]     = equity_capital[t] / face_value[t]
   E[t]          = raw_close[t] * shares[t]
   D[t]          = borrowings[fy] * debt_fraction
```

Bonuses are deliberately excluded from that product. Counting one twice - once in the price adjustment and again through the higher equity capital - would halve the market capitalisation before the event. Splits must be counted, because equity capital does not move. Genuine issuance comes through equity capital, so Yes Bank's fivefold FY2021 dilution is tracked rather than mistaken for a price collapse.

The fix is verifiable on both cases. Yes Bank's market capitalisation is now continuous across its September 2017 split (Rs 83,988 crore in July against Rs 79,800 crore in September, with no artificial fivefold step) and its equity volatility sits at a realistic 28-30% rather than 170%. HDFC Bank's is continuous across its September 2019 split at roughly Rs 6.8 lakh crore.

The debt figure and share count for any date come from the most recent fiscal year already public on that date, on the same 30 June rule as Analysis 1.

24. Parameters and their defaults

Parameter	Default	Note
Horizon T	1 year	Standard KMV convention
Risk-free rate r	6.5%	Approximate RBI repo over the window; a slider in the tool
Default point D	1.0 x borrowings	KMV uses short-term debt plus half of long-term. The archive reports only total borrowings, so this scales it; also a slider
Volatility window	250 trading days	Minimum 120 observations inside the window
Annualisation	sqrt(252)	Daily returns; the solver is evaluated on a weekly grid

Because the default point and risk-free rate are genuinely uncertain, the tool shows a sensitivity grid across both. If the ordering of institutions survives those shifts, the ranking is trustworthy even where the level is not.

25. What it produces

153 institutions solved, roughly 44,900 weekly distance-to-default and implied-PD points from June 2013 to December 2023. For Yes Bank the series does what the original deck only illustrated with invented numbers:

Date	Price	Market cap (Rs cr)	Equity vol	DD	Implied 1y PD
Jul 2017	1,841.8	83,988	30%	3.99	0.0%
Sep 2017	350.0	79,800	28%	4.23	0.0%
Aug 2018	373.2	86,023	28%	4.07	0.0%
Sep 2018	183.7	42,331	48%	2.20	1.4%
Jan 2019	219.6	50,618	58%	1.75	4.0%
May 2019	141.2	32,547	75%	1.17	12.1%
Sep 2019	68.5	15,869	88%	0.77	22.0%
Jan 2020	47.1	10,904	98%	0.56	28.8%
Mar 2020	16.1	3,739	128%	-0.04	51.5%

Two things to read off this. The price falls fivefold between July and September 2017 while market capitalisation barely moves - that is the 5:1 split, correctly handled. And the inflection is **September 2018**, which is when the RBI refused to extend the chief executive's term; the model had no knowledge of that event and picked it up from

price and volatility alone.

The moratorium was 5 March 2020. The accounts-based index only reached 69 on the FY2019 filings; the market was pricing roughly a one-in-three chance of default four months before the event, and had been above 10% since mid-2019. **This is the clearest evidence for running both analyses**, and the comparative page of the tool is built on exactly that disagreement.

PART IV

Validation

How the test was constructed, what it returned, and why the conventional accuracy metric gives the wrong answer on this event set.

26. The event set

Eleven events, each with a date and public source, recorded in `events.csv`. Three are hard credit events; eight are RBI Prompt Corrective Action placements, which are supervisory judgements of distress with known dates and so serve as genuine labels. Searches for Lakshmi Vilas Bank, SREI and DHFL confirmed they are absent from the archive entirely - see Limitations.

27. Lead-time results

Eighteen events. A teal dot marks the seven institutions recovered from source in Part V - the frozen model had never seen any of them.

Institution	Event	Date	First alert	Lead
Indian Overseas Bank	PCA	Oct 2015	FY2012	39 mo
Dhanlaxmi Bank	PCA	Nov 2015	FY2012	40 mo
United Bank of India ●	PCA	Jan 2017	FY2013	42 mo
UCO Bank	PCA	May 2017	FY2012	58 mo
Dena Bank ●	PCA	May 2017	FY2016	10 mo
IDBI Bank	PCA	May 2017	-	**missed**
Central Bank of India	PCA	Jun 2017	FY2012	59 mo
Bank of Maharashtra	PCA	Jun 2017	FY2012	59 mo
Oriental Bank of Commerce ●	PCA	Oct 2017	FY2017	3 mo
Bank of India	PCA	Dec 2017	FY2016	17 mo
Corporation Bank ●	PCA	Dec 2017	FY2014	41 mo
Allahabad Bank ●	PCA	Jan 2018	FY2015	30 mo
Religare Enterprises	default	Jan 2019	FY2013	67 mo
Reliance Home Finance	default	Sep 2019	FY2019	2 mo
Lakshmi Vilas Bank ●	PCA	Sep 2019	FY2013	75 mo
Yes Bank	moratorium	Mar 2020	FY2019	8 mo
SREI Infrastructure Finance ●	resolution	Oct 2021	FY2021	3 mo
Reliance Capital	resolution	Nov 2021	FY2018	41 mo

Alerted before the event: **17 of 18**. Alerted inside the 24-month actionable window: **14 of 18**. IDBI Bank remains the only institution never flagged. Leads beyond about three years should be read as standing concern rather than timed warning.

28. Separation, and the AUC trap

Signal	AUC	Persistence	Permanently flagged 8+ yrs	of which failed
Fundamental Distress Index	0.900	38%	3	0
Borrowings / equity alone	**0.976**	62%	22	**0**
Equity / total assets alone	0.932	-	-	-
Return on assets alone	0.762	-	-	-

AUC is the Mann-Whitney probability that a randomly chosen pre-event company-year scores worse than a randomly chosen other one. Persistence is the share of a flagged institution's observed years spent on the alert list.

Two cautions. Equity/total-assets and borrowings/equity are the **same ratio inverted** (Spearman -0.86), so they are one baseline reported twice, not two independent confirmations. And the event set is clustered: with 13 of 18 events being thin-capital banks under PCA, a capital ratio is partly just detecting ownership type.

The decisive column is the last. Leverage alone permanently flags 22 institutions for eight years or more and **none of them ever failed**. It is a classification of business model, not a warning. **AUC is the wrong headline metric for an early-warning system on a clustered event set;** lead time, persistence and watchlist size are what matter.

29. Cost of the alert list, and controls

Universe averages 502 institutions per year, of which 112 are material.

Rule	Caught	Names / yr	Material names / yr
Level only: FDI ≥ 70	16 / 18	82	34
Level only: FDI ≥ 80	13 / 18	32	12
Level only: FDI ≥ 90	4 / 18	11	3
Level OR deterioration (used)	**14 / 18**	**54**	**15**

Six institutions that never came close to distress, tracked FY2013-FY2023: Kotak Mahindra Bank 27-35, HDFC Bank 30-43, Axis Bank 36-62, ICICI Bank 38-59, Bajaj Finance 52-70, State Bank of India 63-76. The highest reading any control produced was 76, below the alert line.

30. Components removed after testing

Removed component	AUC	Why it fails
Growth burst *2-year borrowings CAGR*	0.27	The intuition runs backwards at this horizon. A lender still growing its book is one wholesale markets still fund; firms about to fail are already shrinking.
Funding stress *operating cash flow / assets*	0.42	For a lender, operating cash flow is dominated by loan-book movement, so it is noise rather than a liquidity read.

Both scored below a coin flip in the PSU-bank cluster and the idiosyncratic-failure cluster independently, so the sign is consistent rather than small-sample noise. They were removed rather than down-weighted.

PART V

Pre-Mortem

Working backwards from the assumption that the system has already failed. Three experiments, each aimed at one of the ways that is most likely to happen. All three are reproducible: python3 premortem.py.

31. The two ways this fails

False negatives: it misses a distressed institution, or flags it too late to act. **False positives:** it flags so many institutions, so persistently, that nobody reads the list any more. The second failure is the more likely one, and the leverage baseline in Part IV is a worked example of what it looks like.

32. Experiment 1 - the unseen-event test

Risk: the two dropped components and the alert thresholds were chosen while looking at every event then available, so the backtest may be measuring the selection rather than the model.

Test: redo the component decision using only the eleven events before 2018 and only accounts up to FY2016, freeze everything, then run it on the seven later events it has never seen.

Component	AUC on development data only	Decision
capital_buffer	0.883	keep
leverage	0.878	keep
profitability	0.809	keep
interest_burden	0.775	keep
capital_erosion	0.705	keep
earnings_trend	0.673	keep
funding_stress	0.417	**drop**
growth_burst	0.270	**drop**

Result: the kept set is **identical** to the shipped model. The drop decision replicates without ever seeing the later events, so it is not an artefact of hindsight. The frozen model then caught **7 of 7** held-out events, with a watchlist of 66 names a year (21 material):

Held-out event	Type	First alert	Lead
Lakshmi Vilas Bank	PCA	FY2013	75 months
Religare Enterprises	default	FY2013	67 months
Reliance Capital	resolution	FY2018	41 months
Allahabad Bank	PCA	FY2015	30 months
Yes Bank	moratorium	FY2019	8 months
SREI Infrastructure Finance	resolution	FY2021	3 months
Reliance Home Finance	default	FY2019	2 months

What this does not prove. Seven events is still few, and the split is temporal rather than random, so the two halves differ in composition as well as date. It rules out one specific failure - that the result is purely an artefact of the selection step - and nothing more.

33. Experiment 2 - the alert-utility test

Risk: a rule can score well on recall and still be unusable, because it flags structural leverage rather than approaching distress.

Test: score every rule on what an analyst actually experiences - how many names land on the desk, how often an alert is followed by an event, and how many institutions are permanently lit up.

Rule	Caught	Precision	Alerts/yr	Material/yr	Persistence	Always-on
FDI ≥ 80 (level only)	13/18	5.9%	32	12	33%	2
FDI ≥ 70 (level only)	16/18	3.7%	82	34	42%	8
FDI level OR deterioration	**14/18**	3.9%	**54**	**15**	**27%**	**2**
Borrowings/equity, top decile	15/18	6.8%	49	45	62%	22
Equity/assets, worst decile	17/18	7.0%	51	33	60%	19
Merton DD, worst decile	4/18	**7.2%**	**9**	4	23%	**0**

Precision is the share of alert-years followed by an event within 24 months; persistence is the share of a flagged firm's observed years spent on the list; always-on counts firms flagged in every one of their 8 or more observed years.

Every precision figure is low single digits, and that is the honest headline. With 18 events among roughly 500 institutions the base rate is about 0.4%, so even a good screen produces mostly non-events. These numbers are only usable relative to each other.

Read that way, two things stand out. The single-ratio baselines keep 19 and 22 institutions permanently lit, against 2 for the composite - the same finding as Part IV, now in the units an analyst cares about. And the **market-based model still has the best precision of anything tested (7.2%) on much the smallest watchlist (9 names a year) with no always-on names at all**, while catching only 4 of 18 events. Its watchlist is small partly because it can only rank the ~150 listed institutions, so the figure is not strictly like-for-like.

That is not a defect, it is the division of labour the two-analysis design predicts. Merton only sees listed firms, and it is structurally blind to the public-sector banks that dominate the event set, because their share prices embed an implicit state guarantee and never priced distress. Where it can see, it is sharp and selective. Where it cannot, the accounts are all there is.

34. Experiment 3 - the coverage test, and closing the gap

Risk: the archive may systematically exclude the hardest cases, making the system look more reliable than it is.

Test: enumerate the Indian bank and NBFC distress events of 2013-2023 that a complete study would cover, and check how many are present. The first run of this test was the worst result in the project: **only 11 of 22 were evaluable, and every single missing name was a failure.** The archive keeps survivors and drops institutions that merge, resolve or delist.

That gap has since been closed, not merely documented. The archive's schema is screener.in's balance-sheet layout, so the same statements could be pulled for the missing institutions in identical accounting definitions and dropped straight into the panel - no reconciliation, no re-derivation. Seven of the eleven were recovered this way:

	Count	Institutions
Originally evaluable	11	Yes Bank, Reliance Capital, Reliance Home Finance, Religare, Dhanlaxmi, IDBI, Indian Overseas, UCO, Central Bank, Bank of Maharashtra, Bank of India
Recovered from source	**7**	**Allahabad, Corporation, Dena, Oriental Bank of Commerce, United Bank of India, Lakshmi Vilas Bank, SREI Infrastructure Finance**
Still unrecoverable	4	DHFL (delisted, statements withdrawn); IL&FS and Altico Capital (never listed); Punjab & Maharashtra Co-op (outside the listed universe)

Coverage rises from **11 of 22 (50%) to 18 of 22 (82%)**, and the panel grows to 563 institutions and 6,223 company-years. Every recovered institution is a failure, so this removes the most damaging part of the bias rather than a random slice of it.

The recovered institutions were scored by a model frozen before they existed in the panel - same components, same weights, same thresholds, no retuning of any kind. On the seven it had never seen, the frozen rule alerted inside the 24-month window for six:

Recovered institution	Event	Date	Alerted in 24m	Lead
United Bank of India	PCA	Jan 2017	yes	18 months
Dena Bank	PCA	May 2017	yes	10 months
Oriental Bank of Commerce	PCA	Oct 2017	yes	3 months
Corporation Bank	PCA	Dec 2017	yes	17 months
Allahabad Bank	PCA	Jan 2018	**no**	- (peak 81)
Lakshmi Vilas Bank	PCA	Sep 2019	yes	15 months
SREI Infrastructure Finance	resolution	Oct 2021	yes	3 months

Adding seven banks moved the existing peer groups only marginally - the Banks group goes from 39 to 41 in FY2017, and the control range for HDFC Bank shifts from 30-43 to 28-43, Kotak from 27-35 to 25-32. The ranking is not being rewritten by the new entrants.

What remains. Four events are still unevaluable and all four are failures, so measured recall stays somewhat optimistic. DHFL is the painful one: a large housing-finance default whose statements have been withdrawn from the source, and recovering it would need annual reports transcribed by hand.

35. What would change the verdict

The question worth holding the team to: *if an Indian NBFC defaults tomorrow, what evidence would show the model genuinely had a chance to see it coming?* Concretely:

- The institution was in the scored universe at the time, with accounts no more than 18 months stale.
- Its index was already elevated, or had moved sharply against its own history, on the last filing published before the event.
- It appeared on a watchlist of a size someone could actually work through - a few dozen names, not several hundred.
- If listed, its distance to default had been falling for more than one quarter.
- And the contrary evidence: if the alert had been continuously on for five years, the model did not predict anything. It described a weak institution, which is a different and much cheaper claim.

36. Experiment 1, corrected - the contamination I had missed

Experiment 1 as first run re-selected the *components* on pre-2018 evidence but kept the alert *thresholds* that were chosen while all events then available were visible. That matters specifically because the deterioration rule was designed around Yes Bank's FY2019 score of 69 - and Yes Bank sits in the held-out set. The original held-out claim was therefore not clean.

Re-deriving the thresholds on development events alone, under a selection rule fixed before searching (maximise development events caught, tie-break on the smaller material watchlist):

	Shipped thresholds	Re-derived on development data only
Level trigger	FDI ≥ 80	FDI ≥ 70
Deterioration trigger	FDI ≥ 60 and jump ≥ +15	FDI ≥ 65 and jump ≥ +25
Development events caught	-	10 of 11
Material watchlist	15 names/yr	35 names/yr
Held-out events caught	7 of 7 (contaminated)	**7 of 7 (clean)**

The re-derived thresholds are **different** from the shipped ones, and they still catch all seven held-out events - Yes Bank at 8 months, now through a rule that never saw it. That is a stronger result than the original claim, and an honest one. It comes at a real cost: the development-derived rule is looser and more than doubles the material watchlist, from 15 names a year to 35.

37. The permutation test - could this be chance?

With eighteen events among roughly 500 institutions, the question that matters is not AUC or recall but whether the result could arise by luck. The null hypothesis is that the alert rule carries no information about *which* institution failed.

The null keeps the real event dates and the real sector of each event, and permutes only which institution failed, drawn from institutions of the same sector actually scored in that window. Scattering events uniformly across the panel would make the test far too easy to pass. 4,000 permutations.

Statistic	Observed	Null mean	Null 95th pct	p-value
Events alerted (all 18)	17	7.20	11.0	< 0.00025
Alert precision	2.55%	1.02%	1.55%	< 0.00025
Events alerted (held-out 7 only)	**7**	**2.18**	-	**0.0003**

The first two rows use a rule whose thresholds were derived on 11 of those 18 events, so they are partly in-sample. **The third row is the clean one:** on the seven events the thresholds never saw, catching all seven has a probability of 0.03% under the null. Under chance, a random same-sector institution is alerted before its assigned event about 2.2 times out of 7.

This is the strongest statistical statement the project can make, and it is worth being precise about what it does and does not establish. It rules out the hypothesis that the alert rule is picking institutions at random. It says nothing about the four distress events still missing from the panel, and a permutation test cannot repair a biased sample.

38. Intervals and event types

Every headline figure with an interval attached, so that no two of them get compared as though the difference were real:

Measure	Estimate	95% confidence interval
Overall recall within 24 months	14 of 18 (78%)	52% to 94%
Median lead time	12 months	6 to 17 months
Recall, PCA placements	9 of 13 (69%)	39% to 91%
Recall, credit events	5 of 5 (100%)	48% to 100%
Held-out recall (Experiment 1)	7 of 7 (100%)	59% to 100%

Recall intervals are Clopper-Pearson; the lead-time interval is a bootstrap over 5,000 resamples.

Two things follow. First, the overall recall interval now **excludes 50%**, which it did not before the seven institutions were recovered - that is the single clearest benefit of closing the coverage gap. Second, the PCA and credit-event intervals still overlap heavily, so the gap between 69% and 100% is not evidence that the model handles credit events better.

The same logic retires several comparisons made earlier in this report: the precision figures in section 33 sit inside each other's error bars and should not be read as one rule beating another.

39. Testing five more signal families

Both existing analyses share a blind spot: one reads the accounts, the other needs a liquid listed share, and neither sees funding pressure or earnings quality directly. Further signal families were built and tested, each as a separate module scored on the same peer-relative scale, at the same point-in-time date, against the same 18 events.

Three could not be built at all, and it is worth being explicit about why: **credit-rating actions** have no free, consistent historical feed of Indian rating changes with dates; **bond and CDS spreads** are not publicly available at instrument level for Indian issuers; **trading volume** was discarded when the price extracts were cached and would need a full re-fetch.

Signal	Coverage	AUC (to FY2016)	AUC (FY2017+)
Market: drawdown from peak	20%	0.631	0.777
Market: abnormal 12-month return	17%	0.533	0.769
Market: volatility spike	17%	0.478	0.599
Funding: loan book shrinking	**61%**	**0.736**	**0.624**
Funding: financing cash outflow	98%	0.566	0.536
Funding: thin liquid buffer	100%	0.390	0.534
Quality: profit minus operating cash	98%	0.321	0.438
Quality: other-income reliance	97%	0.299	0.496
FDI composite, for reference	*100%*	*0.906*	*0.892*

Only one of the eight has both a usable AUC and enough coverage to matter: a shrinking loan book. The market signals discriminate well in the later period but exist for only a fifth of the panel.

40. Do they add anything? Mostly not

Discrimination is not the test. The test is whether adding a signal to the alert rule catches events the index misses, warns earlier, or shrinks the watchlist. Each row adds one signal as an OR condition firing on its worst decile:

Added signal	Events caught	Names/yr	Material/yr
Baseline: FDI alert rule	*14 / 18*	*54*	*15*
+ Market: drawdown	14 / 18	63	19
+ Market: abnormal return	14 / 18	60	19
+ Market: volatility spike	14 / 18	62	20
+ Funding: loan book shrinking	**15 / 18**	**85**	**18**
+ Funding: financing outflow	14 / 18	97	23
+ Funding: thin liquid buffer	14 / 18	94	26
+ Quality: profit minus cash	14 / 18	100	26
+ Quality: other-income reliance	14 / 18	98	23

Seven of the eight add nothing and cost a great deal. They catch no additional event while enlarging the watchlist by between 10% and 85%. The eighth, loan-book contraction, buys one extra event for 31 more names a year - worth noting, not obviously worth making. **None of these signals has been added to the model.**

Two findings are worth more than the negative result itself. First, **earnings-quality measures are inverted for lenders.** The profit-minus-cash gap scores 0.32, well below chance, because a lender's operating cash flow is dominated by loan-book movement: a shrinking book releases cash, so distressed lenders show cash above profit, not below. The corporate-finance intuition does not transfer, and the same mechanism explains why funding-stress was dropped from the index in the first place.

Second, the signals are not independent of the index in the way the architecture assumed. The institutions a drawdown signal flags are largely the ones the accounts already condemn; the market adds timing, which Analysis 2 already captures, rather than new names. That is an argument for keeping Merton as a separate instrument and against folding market data into the fundamental score.

On the four events the index misses - IDBI, Central Bank of India, Bank of Maharashtra and Allahabad Bank, all PCA placements in 2017-18 - loan-book contraction reaches the 92nd and 85th percentile for Central Bank and Allahabad, but only the 71st and 47th for IDBI and Bank of Maharashtra. No signal tested catches all four, and none catches IDBI.

PART VI

External Validation on US Bank Failures

The Indian study is limited by events, not data. The only way to test whether the methodology generalises is to run it, unchanged, where the event population is large.

41. Why this test, and why the United States

Eighteen events cannot settle whether a method works. The FDIC publishes, free and without authentication, every US bank failure since 1934 and quarterly call-report financials for every insured institution, joined on a single identifier. That gives three things the Indian data cannot:

- **Events at scale.** 547 evaluable failures against 18.
- **Quarterly frequency.** The index can move four times a year instead of once.
- **No survivorship bias.** Failed banks remain in the data with the call reports they filed before failing — the exact defect that required recovering seven Indian institutions by hand.

Nothing was refitted. Same six components, same weights, same dual anchoring, same peer-relative percentiles, same alert rule (80 / 60 / +15), same 24-month horizon, same point-in-time discipline. Only the field names change, because a call report and a balance sheet record the same quantities under different labels.

Component	India	United States
capital_buffer	equity / total assets	EQ / ASSET
capital_erosion	2-year change in equity	8-quarter change in EQ
profitability	net profit / total assets	ROA
leverage	borrowings / equity	(ASSET – EQ) / EQ
interest_burden	interest expense / revenue	EINTEXP / INTINC
earnings_trend	3-year change in ROA	12-quarter change in ROA

Field mapping. The model is unchanged; only its inputs are renamed.

42. Scale of the test

	India	United States
Panel	6,223 company-years	**416,285 bank-quarters**
Institutions	563	**10,341**
Period	FY2006–FY2023, annual	2003–2015, quarterly
Evaluable events	18	**547**
Pre-event observations	36	**4,309**

43. Result: the method transfers

Measure	Value
Failures alerted within 24 months	**535 of 547 (98%)**

Measure	Value
Median lead time	**19.6 months** (quartiles 13.1 / 22.5)
Alert precision	7.32% against a 1.04% base rate — **7.1x lift**
Watchlist	829 of 8,005 banks per quarter (10%)

A 98% recall is bought with a watchlist of a tenth of the banking system, which is the honest cost and would be too wide for some users. The discriminating figure is the 7.1x lift over the base rate, and the permutation result below.

44. The AUC trap, settled

Part IV reported an uncomfortable finding: on the Indian data a single leverage ratio scored a *higher* AUC than the composite (0.976 against 0.912), and the defence was that the event set was clustered on thin-capital public-sector banks. That defence is testable here, where no such clustering exists.

Signal	AUC	Caught	Persistence	Always-on
Distress index composite	**0.861**	**535 / 547**	**19%**	**2**
Leverage alone	0.828	511 / 547	27%	27

On 547 events the composite beats leverage on every measure, including AUC. The Indian ranking was an artefact of a clustered event set, exactly as argued, and leverage alone remains the worse alarm: 27 institutions permanently flagged against 2.

45. Permutation test at scale

Statistic	Observed	Null mean	Null 95th pct	p-value
Failures alerted	**535**	141.9	160	< **0.001**

The null preserves failure dates and size terciles, permuting only which bank failed, over 1,000 draws. Chance assigns an alert to roughly 142 of 547; the rule finds 535. This is the same test as Part V, on thirty times the events.

46. What it reveals about the Indian data

The call reports carry a non-performing-assets ratio for every bank-quarter. Tested as a standalone signal it scores **AUC 0.887 at 100% coverage** — stronger than any single component in the model.

The Indian archive carries the same quantity only from 2020 and for a tenth of the panel, which is why it was excluded from that backtest. **The implication is that the principal weakness of the Indian system is a missing input, not a flawed structure.** Obtaining a historical asset-quality series for Indian lenders — from quarterly exchange filings or RBI statistical tables — would plausibly do more for performance than any further modelling.

What this does not establish. US and Indian banking differ in regulation, funding structure and resolution practice, and the US window is dominated by one crisis. Transferring well to US failures is evidence that the method is not an artefact of the Indian sample; it is not evidence that the Indian results themselves are precise.

47. Building it from scratch on US data

Section 41 asked a narrow question: does the frozen model transfer? It does. A different question is what the same design produces when the data is good enough to build on. The panel was extended to **588,177 bank-quarters across 10,459 institutions, 2003-2023**, and three components the Indian archive could not support were added:

- **Non-performing assets ratio.** Reported every quarter for every bank. The Indian archive carries it only from 2020, for a tenth of the panel.
- **Reserve shortfall** — bad loans not covered by loss reserves. Measures whether a bank has recognised the losses it already has.
- **Deposit contraction.** Deposits are reported quarterly, so funding flight can be measured directly rather than proxied by a shrinking loan book.

The split is across **two different crises**, not merely two date ranges: components and thresholds are derived on the 529 failures before 2013 (the 2008-12 wave) and tested on the 68 from 2013-2023, which includes the 2023 regional bank failures.

48. Component selection on 529 development failures

Component	AUC on development data
Non-performing assets ratio *	**0.875**
Reserve shortfall *	**0.856**
Profitability	0.830
Leverage	0.815
Capital buffer	0.814
Earnings trend	0.783
Interest burden	0.779
Capital erosion	0.768
Deposit contraction *	0.580

* unavailable in the Indian data. All nine clear the 0.5 bar, so none is dropped — and the two asset-quality measures are the strongest signals available anywhere in this project.

49. A flaw in the selection rule, caught and fixed

The first run selected thresholds by maximising development recall, breaking ties on watchlist size. With 529 events that rule loosens the threshold until almost everything is flagged: it chose `FDI ≥ 70` and produced a watchlist of **18% of the banking system** — precisely the failure Part IV criticises in the leverage baseline.

The rule was replaced with one stated before searching: **maximise recall subject to a watchlist no larger than 5% of banks**. That is a judgement about what an analyst can work through, and it belongs in the selection rule rather than in the commentary afterwards.

Selection rule	Thresholds	Held-out watchlist	Held-out precision
Maximise recall	70 / 50 / +10	18% of banks	0.74% (4.7x lift)
Recall subject to 5% cap	**85 / 65 / +20**	**2.8% of banks**	**3.93% (24.8x lift)**

50. Held-out result, and what better data actually buys

Measure	Value
Failures alerted within 24 months	**55 of 68 (81%)**

Measure	Value
Median lead time	14.5 months (quartiles 8.2 / 22.1)
Watchlist	159 of 5,675 banks per quarter (2.8%)
Precision	3.93% against a 0.16% base rate — **24.8x lift**
Permutation test	observed 55, null mean 4.6, **p < 0.001**

Tested on a regime it was not developed on, the model catches four failures in five with a list of under 3% of banks, at roughly twenty-five times the base rate. Under chance, the same list would have caught about five.

The most useful comparison is what the three new components changed:

Model	Caught	Median lead	Precision	AUC
Six components, as in India	55 / 68	14.4 months	2.04%	0.895
Nine, with asset quality and funding	**55 / 68**	14.5 months	**3.93%**	0.894

Identical recall, identical lead time, identical AUC — and half the false alarms. Asset quality is the strongest standalone signal in the project (AUC 0.875), yet adds nothing to *which* institutions are caught, because capital and profitability already identify them. What it adds is discrimination among the ones that look weak: it halves the list needed to catch the same failures.

That is a precise answer to what better Indian data would be worth. Not more events detected — a watchlist half the size for the same detection. On the Indian panel, where the material watchlist runs to 14 names a year, that is the difference between a list someone reads and a list someone ignores.

PART VII

Limitations

Read this before quoting any number in this report. Several are serious enough to change how the output should be described.

51. Statistical

- **Eighteen events is still a small sample.** Every Indian figure carries wide error bars; the external validation in Part VI exists precisely because 18 events cannot settle whether the method works.
- **Component selection used the same events it was tested on.** Dropping two components after observing their performance is a mild form of fitting. There is no held-out set, so reported accuracy is optimistic.
- **The event set is clustered.** Thirteen of eighteen events are banks placed under PCA within a 26-month span, so the test leans heavily on one failure mode in one part of the cycle.
- **Peer groups are small.** 39 banks means one institution moves roughly 2.5 percentile points by existing, and NBFC size terciles are recomputed annually, so a firm can change group - and score - without changing itself.

52. Interpretation

- **The fundamental score is a rank, not a probability.** An 88 means “worse than 88% of its peers”, not “88% likely to default”. The original project deck frames the output as Probability of Default; that wording should change, because readers will take a percentage as a likelihood.
- **The Merton probability is a model output, not a forecast.** It assumes lognormal asset value, a single debt maturity and no external support. Its comparative movement is trustworthy; its level is not.
- **A high score is not a prediction of failure.** Most institutions scoring above 80 will not fail. The claim is narrower: institutions that failed were usually scoring high beforehand.
- **Long lead times are not a success.** A 59-month warning on a bank that scored badly every year is a standing concern, not a timed signal.

53. Data

- **Analysis 1 can only see what the accounts admit, and this is the deepest limitation.** Yes Bank's reported profitability ranked mid-pack to the end because the bad loans were not yet recognised. A model built on filings inherits whatever those filings understate or conceal. Analysis 2 exists because of this.
- **The archive deletes its own hardest cases.** Institutions that merge or resolve leave it. Seven were recovered from source filings (Part V), but four failures remain unevaluable - DHFL, IL&FS, Punjab & Maharashtra Co-op and Altico - and all four are failures, so measured recall stays optimistic.
- **Annual accounting data only.** The fundamental index updates once per filing cycle at best. Quarterly NPA data begins only in 2020 and covers about a tenth of the panel, so it is shown as current context and excluded from the backtest.
- **Volatility is a trailing estimate.** The 250-day window smooths genuine regime changes, so the market signal lags a sudden repricing by weeks.
- **Accounting quality is not verified.** Reserves are taken as reported and may include revaluation and other non-loss-absorbing items.
- **The snapshot ends in 2023.** Nothing after that is covered.

54. Scope

- **Analysis 2 covers only listed institutions with a liquid quote** - 152 of 556. It cannot be applied to unlisted NBFCs, and should not be faked for them.
- **No notion of sovereign or parent support.** IRFC and LIC Housing score in the 80s on thin equity and high leverage; both are government-linked and were never near distress. Merton makes the same mistake in the other direction for public-sector banks, whose market prices embed an implicit state guarantee.
- **Group and related-party exposure is invisible.** The mechanism that actually brought down IL&FS and Reliance Capital - lending into a connected web of subsidiaries - does not appear in a standalone balance sheet.
- **Not a credit rating and not investment advice.** An academic decision-support exercise built entirely from public data.

55. Reproducing this

```
pip install pandas numpy streamlit jinja2

python3 build_panel.py        # archive  -> panel.csv   (6,139 rows)
python3 score.py              # index    -> scored.csv
python3 validate.py           # backtest -> lead_times.csv
python3 fetch_nse_prices.py   # NSE      -> prices_weekly.csv
python3 resolve_splits.py     # actions  -> splits.csv
python3 merton.py             # model    -> merton.csv
python3 premortem.py          # the three experiments in Part V
python3 validation2.py        # thresholds, permutation test, intervals
python3 recover_missing.py    # the 7 institutions the archive dropped
python3 signals.py            # the five extra signal families in Part V
python3 fdic_fetch.py         # US failures + call reports -> fdic_panel.csv
python3 fdic_validate.py      # the frozen model on 547 US failures (Part VI)
python3 fdic_model.py         # the US-native build, 597 failures, two regimes
python3 recover_missing.py    # the 7 institutions the archive dropped
python3 signals.py            # the five extra signal families in Part V
python3 fdic_fetch.py         # US failures + call reports -> fdic_panel.csv
python3 fdic_validate.py      # the frozen model on 547 US failures (Part VI)
python3 fdic_model.py         # the US-native build, 597 failures, two regimes

streamlit run app.py          # the interactive tool
```

Every number in this report is produced by those scripts from the archive and NSE's public price files.